Major Project Abstract

Project Title

Agentic Desktop Assistant using Telegram and Local AI

Abstract

The proposed project is an Agentic Desktop Assistant that enables users to securely interact with and control their personal computer remotely through Telegram using natural language commands.

Unlike traditional remote desktop applications that require continuous manual operation, the proposed system interprets user requests, plans the required sequence of actions, executes them on the desktop, and provides progress updates back to the user.

The assistant is designed around a modular architecture consisting of a planning module, desktop execution module, communication module, and task management module. This modular design allows future expansion without changing the core architecture.

The system focuses on improving productivity by automating routine desktop tasks such as application launching, file management, browser interaction, and basic workflow execution while maintaining a simple and extensible design.

Problem Statement

Existing desktop automation and remote access solutions have several limitations:

- They primarily depend on manual remote control.
- Most automation tools require predefined scripts instead of natural language.
- Existing assistants have limited task planning capabilities.
- Users often need to manually perform repetitive desktop operations.
- Many systems are difficult to extend with new capabilities.

Proposed Solution

Develop an intelligent desktop assistant capable of:

- Receiving commands through Telegram.
- Understanding user intent.
- Planning simple multi-step workflows.

- Executing desktop tasks automatically.
- Providing execution status and feedback.
- Maintaining a modular architecture for future expansion.

Major Modules

Communication Module

- Telegram Bot Interface
- Command handling
- User authentication

Planning Module

- Interpret user requests.
- Convert requests into executable task sequences.
- Manage task flow.

Desktop Execution Module

- Open applications.
- Manage files and folders.
- Perform desktop automation.
- Execute supported workflows.

Task Management Module

- Track execution progress.
- Report task status.
- Handle basic failures gracefully.

Expected Outcomes

- Intelligent desktop task automation.
- Natural language interaction.
- Modular and extensible architecture.
- Improved productivity for repetitive desktop tasks.
- Foundation for future intelligent desktop systems.

Novelty of the Project

Unlike conventional desktop automation tools that mainly execute predefined commands or scripts, the proposed system introduces an agent-oriented workflow where user requests are interpreted, planned, and executed through modular components. This enables more flexible task execution while providing an architecture that can be extended with additional intelligent capabilities in future versions.

Scope

Included

- Telegram-based interaction.
- Natural language task handling.
- Multi-step task execution.
- Desktop automation.
- Modular software architecture.
- Progress reporting.

Future Scope (Not Part of Current Project)

- Voice interaction.
- Multi-agent collaboration.
- Long-term memory.
- Advanced coding assistant.
- Autonomous learning.
- Vision-based desktop understanding.
- Additional communication interfaces.